Example 5.21 Expand $(1+x)^{\frac{2}{3}}$ up to four terms for $|x|<1$.

Solution:
Here $n=\frac{2}{3}$.

$$\begin{aligned}
\frac{n(n-1)}{2!} &= \frac{\frac{2}{3}(\frac{2}{3}-1)}{2!} \\
&= \frac{\frac{2}{3}(\frac{-1}{3})}{2} \\
&= \frac{-1}{9} \\
\frac{n(n-1)(n-2)}{3!} &= \frac{\frac{2}{3}(\frac{2}{3}-1)(\frac{2}{3}-2)}{3!} \\
&= \frac{\frac{2}{3}(\frac{-1}{3})(\frac{-4}{3})}{6} \\
&= \frac{4}{81}
\end{aligned}$$

Thus

$$(1+x)^{\frac{2}{3}} = 1+\frac{2}{3}x-\frac{1}{9}x^2+\frac{4}{81}x^3+\cdots$$

Example 5.22 Expand $\dfrac{1}{(1+3x)^2}$ in powers of x. Find a condition on x for which the expansion is valid.

Solution:
If we take $3x=y$, then

$$\frac{1}{(1+3x)^2}=\frac{1}{(1+y)^2}.$$

Now $\dfrac{1}{(1+y)^2}$ can be expanded using binomial theorem in powers of y. The expansion is valid only for values of y satisfying $|y|<1$. Replacing y by $3x$ we can get an expansion of $\dfrac{1}{(1+3x)^2}$. The expansion is valid only for values of x satisfying $|3x|<1$; that is the expansion is valid only for values of x satisfying $|x|<\frac{1}{3}$.

$$\begin{aligned}
\frac{1}{(1+3x)^2} &= (1+3x)^{-2} \\
&= 1-2(3x)+\frac{2(2+1)}{2!}(3x)^2-\frac{2(2+1)(2+2)}{3!}(3x)^3 \\
&\quad +\frac{2(2+1)(2+2)(2+3)}{4!}(3x)^4-\cdots
\end{aligned}$$

Hence, $\frac{1}{(1+3x)^2}=1-6x+27x^2-108x^3+405x^4-\cdots, |x|<\frac{1}{3}$.